

OLUWAGBEMINIYI REST JOHNSON

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SUMMARY OF QUALIFICATIONS

PhD student in Computing and Informatics (Software & Information Systems) with a background in mathematics and CS. Expertise in machine learning, statistical analysis, and software development. Experienced in integrating quantitative and qualitative methods with advanced data handling to analyze datasets, solve problems, and inform user-focused product decisions. Recognized for bridging research and practice through innovative projects and relevant publications. Research interests focus on human-AI interactions in the context of career development.

EDUCATION

Doctor of Philosophy, Ph.D.

Computing and Informatics, GPA: 3.80

Concentration: Soft. & Inf. Systems
UNC Charlotte

Research Focus: **Human-Centered
Intelligent Systems, Machine
Learning, and Computing Education**

Anticipated Graduation: May 2027

Master of Science, M.Sc.

Inf. Technology, GPA: 3.80

Concentration: **Advanced Data and
Knowledge Discovery**

UNC Charlotte

Graduated: Dec 2025

Bachelor of Science, B.Sc.

Mathematics, GPA: 3.92

Newberry College; Newberry, SC

Graduated: May 2023

CERTIFICATIONS

IBM Agentic AI with LangChain and LangGraph, Jan 2026

Building AI Agents with MongoDB (Credly), Fall 2025

COMPETENCIES

Data Analysis: Python (Pandas, NumPy), MATLAB

AI and Machine Learning: Azure AI, Sci-kit, PyTorch, TensorFlow, LangChain, PyDantic, AG2, CrewAI, OpenAI Vision API, MongoAI, Prompt Engineering

Human Centered Computing: Designing Usability studies and Survey instruments, and Accessible Interfaces

Software Development: JavaScript, Java, MySQL, HTML, CSS. MERN, Django, AWS, MongoDB, Git, Postman

Technical Skills: Quantitative Analysis, Problem Solving, Computational Thinking, Predictive Analytics

Soft Skills: Interpersonal Communication, Ability to explain complex ideas clearly.

RELEVANT PROJECTS & RESEARCH HIGHLIGHTS

Machine Learning for Bug Priority Analysis

- Collected and preprocessed over **24,000** Visual Studio Code bug reports using **Python (Sci-kit, NLTK)**, for NLP model pipelines that reduced noise and enhanced accuracy
- Designed a **BERT-based classification system** for bug triage, which improved model macro F1 by **10%**, and demonstrated complete **bug class identification** compared to other models

Software Design and Development for Problem-Solving

- Co-led a 4-member team through the full software development lifecycle to design **PMaCT** (PhD Milestone and Career Tracker), a Kanban-like PhD Milestone Tracker built on the **MERN** stack.
- Delivered features including **calendar synchronization, automated reminders, and committee communication**, resulting in a platform that streamlined progress tracking for graduate students.

Reinforcement Learning in Cognitive Development

- Designed

- Collected and analyzed **mixed-methods data** for ECS QEM (Exploring Computer Science, Quality Education for Minorities) initiatives, identifying emerging themes and actionable insights for broadening minority CS participation.
- Moderated workshops on **AI in education**, fostering collaboration among **50+ educators** and **researchers** on integrating AI into curriculum design.

Teaching and Research Assistant

| University of North Carolina | Charlotte, NC

August 2023 - Present

- Teaching assistant for Software Engineering, mentoring **120+ undergraduates** on the development and design of sophisticated web applications, APIs, and Microservices using full-stack development frameworks such as **Django** and **MERN**
- Conducting doctoral research on **AI-driven career development systems** that model **machine learning tools** to align student career decisions with industry workforce needs.
- Organized and led Python programming camps for **80+** students, fostering **computational thinking** and **data literacy**, resulting in improved **student understanding** of Python's efficiency and versatility..

Jenzabar Intern

| JENZABAR | Newberry, SC

June 2022 - Aug 2022

- Queried and managed relational databases with **SQL** to generate **1000+** student performance and probation reports, to support the Newberry College registrar workflows and administrative decision-making.

PUBLICATIONS

- **O. Johnson**, M. Mejias, and L. -M. Kouaho, "(WIP) Bridging the Industry-Academia Gap: Assessing the Need for AI Tools in Technical Interview Preparation and Workforce Readiness," 2025 IEEE Frontiers in Education Conference (FIE), Nashville, TN, USA, 2025, pp. 01-05, doi: 10.1109/FIE63693.2025.11328371.
- **O. Johnson**, M. Mejias, L.-M. Kouaho, and V. Mack, "Python Pioneers: A K-12 just-in-time, culturally relevant, computational thinking-based Python learning experience," in Proc. 21st Int. Conf. Frontiers in Education: Computer Science & Computer Engineering (FECS '25), July 21-24, 2025, Las Vegas, USA.
- M. Mejias, **O. Johnson**, A. Mays, L. Burge, and D.-M. Wilson, "Buddy, Not Crutch: Ethical Use of Zero-Shot Prompting with Public LLMs for Software Engineering Technical Interview Preparation," in Proc. Int. Conf. Frontiers in Education: Computer Science (FECS '25), July 21-24, 2025, Las Vegas, USA.
- Marlon Mejias, Dale-Marie Wilson, Eric Betties, **Oluwagbeminiyi Johnson**, Felesia Stukes, Gloria Washington, and Jody Marshall. 2025. Transforming Perceptions of Black Achievement in Computing Research: A Multi-Institutional Intervention. In Proceedings of the 2025 Conference on Research on Equitable and Sustained Participation in Engineering, Computing, and Technology (RESPECT 2025). Association for Computing Machinery, New York, NY, USA, 167–175. <https://doi.org/10.1145/3704637.3734773>

AWARDS

- 2nd Place, Graduate Poster Competition (Tapia 2025)
- Best Paper Award, Experience Report Track (RESPECT 2025)
- George B. Cromer Award (Newberry College 2023): Graduating senior who exemplifies academic excellence, leadership ability, and personal integrity
- Outstanding Achievement in Mathematics (Newberry College 2023)

PROFESSIONAL MEMBERSHIPS AND ASSOCIATIONS

IEEE Graduate Student Member

2025

Pi Mu Epsilon Mathematical Society

2022